

Richie Tran

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EDUCATION

Carnegie Mellon University

Pittsburgh, PA

B.S. Computer Science and Machine Learning

May 2027

Relevant Coursework

Data Structures & Algorithms, Designing Human Centered Software, DSA in Parallel, Computer Systems, AI Representation and Problem Solving, Machine Learning I, Functional Programming

EXPERIENCE

Software Engineer Intern | *Opportunity Knocks*

November 2025 – Present

- Developing a React Native mobile app with Supabase bringing the existing web application to iOS and Android, targeting 4,000+ users at launch.
- Resolved 10+ production bugs and shipped 3 new features across the TypeScript web app and React Native mobile app, including dark mode implementation serving 7,000+ users.
- Implemented an internship builder feature enabling companies to create programs for students, and built transcript management and meeting creation flows using Supabase real-time queries.

Research Assistant | *Reliable Autonomous Systems Lab, CMU*

January 2026 – Present

- Building Victor, a Scrabble-playing agent in C++ integrating computer vision, voice recognition, and OpenAI-powered speech generation for human-robot gameplay with 500 ms response latency.
- Engineering a computer vision pipeline achieving 99% tile detection accuracy across 100 tiles, processing board updates in under 5 seconds per turn.
- Designed an interactive touchscreen Scrabble board in Python simulating realistic physics for 100 tile pieces using PyBullet for human-robot interaction.

Software Developer | *Scotty Labs, CMU*

August 2024 – Present

- Creating a web scraper to aggregate 1,000+ research entries across 100+ CMU labs and departments, enabling students to discover and connect with faculty research opportunities.
- Shipped an AI presentation coach used by 50+ students, analyzing speech pace, filler words, and eye contact in real time via React and Node.js.

PROJECTS

MyEyes | *Python, Swift, Flask, CoreML, YOLOv8*

- Developed an audio-based iOS app enabling visually impaired users to navigate safely through real-time hazard detection, processing 30 frames per second.
- Achieved 90% detection accuracy by deploying a fine-tuned YOLOv8 model via CoreML with a custom urgency-weighted loss function.
- Integrated LiDAR and camera-based obstacle identification with Siri commands for accessibility.

StoryScroll | *Python, JavaScript, React, FastAPI, FFmpeg, ElevenLabs API, Cloudflare R2*

- Created a video generation pipeline converting Reddit stories into short-form content by compositing ElevenLabs TTS with background footage via FFmpeg, generating videos in under 3 minutes.
- Designed an asynchronous job queue with RESTful API endpoints and real-time progress polling, enabling concurrent video processes across a React frontend and FastAPI backend.
- Implemented a Reddit scraping module surfacing the top 30 stories across 5 subreddits with filtering by trending, hot, and new posts, delivering generated videos via Cloudflare R2 for scalable storage.

AI Logo Generator | *Next.js, Express, Claude API, HTML/CSS*

- Built a full-stack web app that extracts brand identity from user-uploaded meeting transcripts, leveraging Claude Sonnet 4.6 for transcript analysis and Opus 4.6 for image generation.
- Architected an Express API pipeline chaining the two models end-to-end, processing raw JSONs into structured brand profiles and producing 5 logos with tailored design rationales in under 3 minutes.

SKILLS

Languages: TypeScript, JavaScript, Python, C, C++, Java, SQL, Swift, HTML/CSS, Bash

Frameworks: React, React Native, Next.js, Node.js, Express, FastAPI, Flask, Tailwind CSS, Bootstrap

Developer Tools: Git, Supabase, Firebase, AWS S3, Cloudflare R2, SQLite, Vercel, FFmpeg, Figma

Libraries/APIs: Claude API, OpenAI, PyTorch, CoreML, YOLOv8, ElevenLabs API, PyBullet